BACHELOR OF PHYSIOTHERAPY

BPT-109 · Computer Assignment–1

Computer Fundamentals, Input & Output Devices

Submission date: 13 October 2026

Name: _____

Roll No.: _____

Assignment overview: Eight questions covering computer characteristics and generations, the functional block diagram, campus scanning systems, programming languages, pointing and drawing devices, blueprint output systems, biometric security, and accessible workstation design.

QUESTION 1 · REMEMBERING

1. Characteristics of computers and five generations

A. Four primary characteristics of a modern digital computer

1. **Speed:** A computer performs millions or billions of calculations in a very short time.
2. **Accuracy:** It produces accurate results when the data and instructions provided are correct.
3. **Storage:** It can store a large amount of data and retrieve it whenever required.
4. **Diligence:** It can perform repetitive tasks continuously without getting tired.

B. Five generations of computers

Generation	Period (approx.)	Main hardware technology	Example
First	1940–1956	Vacuum tubes	ENIAC
Second	1956–1963	Transistors	IBM 1401
Third	1964–1971	Integrated Circuits (ICs)	IBM System/360
Fourth	1971 onwards	Microprocessors using VLSI technology	Personal computers
Fifth	Present and developing	ULSI, AI-oriented processors and parallel processing	AI-based computer systems

> **Conclusion:** Each generation improved computer speed, size, reliability, storage capacity and efficiency.

QUESTION 2 · UNDERSTANDING

2. Functional block diagram of a computer

Definition: A computer block diagram is a graphical representation of the main functional units of a computer and their relationships.

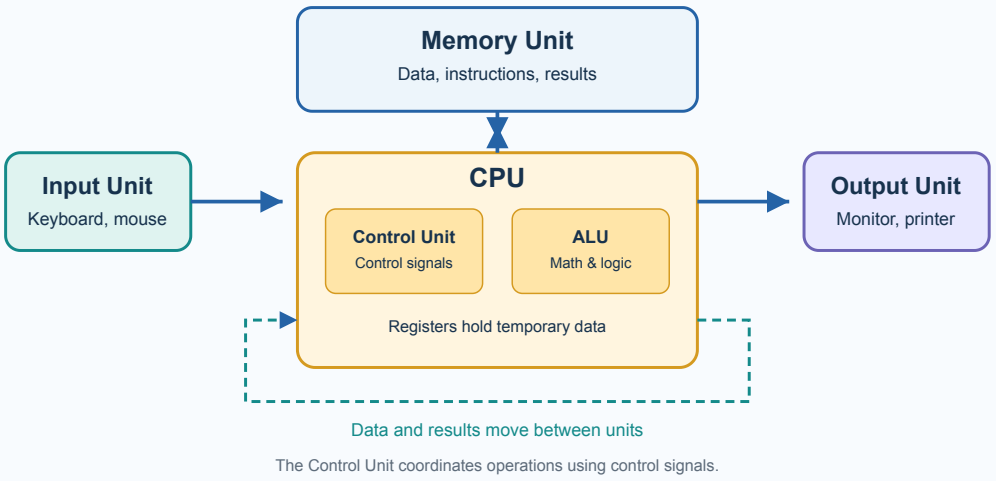


Figure 1. Simplified functional block diagram of a computer.

Main components

- **Input Unit:** Accepts data and instructions through devices such as a keyboard, mouse and scanner, and converts them into a form the computer can process.
- **Memory Unit:** Stores data, instructions and intermediate results. RAM temporarily stores working data, while secondary storage keeps data for longer periods.
- **Central Processing Unit (CPU):** The main processing unit. The **Control Unit (CU)** directs and coordinates operations; the **Arithmetic Logic Unit (ALU)** performs arithmetic and logical operations; **registers** provide fast temporary storage.
- **Output Unit:** Presents processed information through monitors, printers and speakers.

Flow of data and instructions

1. The input unit enters data and instructions.
2. Memory stores the data and instructions.
3. The CPU fetches, decodes and executes instructions.
4. The control unit coordinates operations using control signals.
5. The ALU performs calculations and logical operations when required.
6. Results are stored in memory and sent to the output unit when needed.

> **Conclusion:** The CPU, memory and input/output units work together to convert raw data into meaningful information.

QUESTION 3 · APPLYING

3. Use of scanning devices and card readers on campus

Introduction: Scanning devices and electronic card readers help educational institutions perform daily activities quickly and accurately.

A. Student attendance system

Barcode scanners and Optical Mark Recognition (OMR) scanners can be used for attendance.

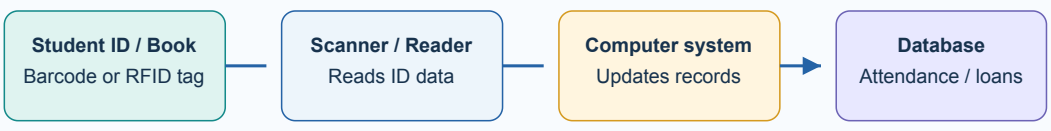
1. Each student receives an identity card containing a barcode.
2. The barcode is scanned when the student enters the classroom.
3. The computer reads the student's identification number.
4. Attendance is automatically recorded in the database.
5. Teachers can view and prepare attendance reports.

OMR can also read marked responses on specially designed attendance sheets.

B. Library checkout system

Radio Frequency Identification (RFID) and smart card readers can be used in libraries.

1. Each student receives an RFID or smart ID card.
2. Each library book is tagged with a barcode or RFID tag.
3. The student's card and the book are scanned or read.
4. The system records the student ID, book details and issue date.
5. The library database is updated and the return date is tracked.



Automated record keeping reduces manual work.

Figure 2. Example campus scanning and record workflow.

Advantages

- Saves time and reduces paperwork.
- Improves accuracy of attendance records.
- Makes book issuing and returning faster.
- Helps prevent duplicate or incorrect entries.
- Makes record keeping easier.

> **Conclusion:** Optical scanners and electronic card readers make campus attendance and library management faster, more efficient and more reliable.

QUESTION 4 · ANALYZING

4. Low-level languages and high-level languages

Definition: Programming languages are used to provide instructions to a computer. Low-level languages are closely related to hardware, whereas high-level languages use instructions that are easier for humans to understand.

Basis	Machine language	Assembly language	High-level language
Meaning	Uses binary digits (0 and 1)	Uses mnemonic codes	Uses human-readable statements
Execution speed	Very fast when executed directly	Generally efficient after translation	Depends on language, compiler and program
Hardware dependence	Highly hardware-dependent	Highly hardware-dependent	Generally more portable
Translation	No separate language translator required	Requires an assembler	Usually requires a compiler or interpreter
Complexity	Very difficult	Difficult	Easier to learn and use
Debugging	Difficult	Difficult	Generally easier
Examples	Binary machine instructions	Mnemonics such as MOV	C, C++, Java, Python

Explanation

Machine language: The lowest-level language, consisting of binary instructions directly understood by the processor.

Assembly language: Uses symbolic instructions instead of binary codes. An assembler translates these instructions into machine code.

High-level language: Uses English-like words and mathematical symbols. A compiler or interpreter translates or executes the program according to the language's implementation.

> **Conclusion:** Low-level languages offer close hardware control, while high-level languages are generally easier to develop, understand and maintain.

QUESTION 5 · ANALYZING

5. Point-and-draw devices compared with graphics tablets

Introduction: Pointing and drawing devices help users select objects, control a pointer and create digital drawings.

Feature	Mouse	Trackball	Graphics digitizer / drawing tablet
Working mechanism	Detects movement across a surface	Stationary device with a ball controlled by fingers or thumb	Detects stylus position on a tablet surface
Coordinate tracking	Relative pointer movement	Relative pointer movement controlled by ball	Precise tablet coordinates, often mapped to screen
Precision	Suitable for general work	Useful for controlled pointer movement	High precision for drawing and technical design
Pressure sensitivity	Usually unavailable	Usually unavailable	Supported by many pressure-sensitive tablets
CAD applications	Selecting tools and objects	Navigation and selection	Tracing and technical drawing
Digital art	Limited for natural drawing	Limited for natural drawing	Sketching, painting and illustration
Cost	Generally affordable	Varies by model	Varies from affordable to expensive

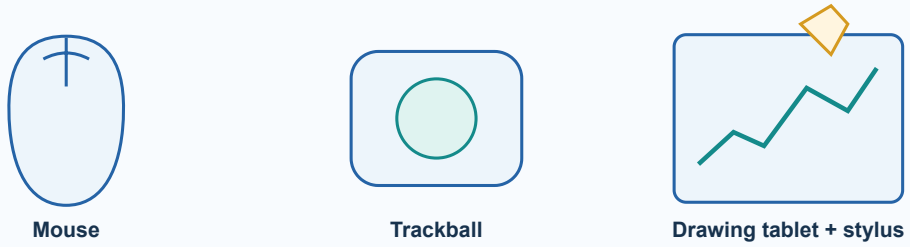


Figure 3. Simplified illustrations of common pointing and drawing devices.

Applications

CAD: A mouse is commonly used to select tools and modify objects. A trackball can help with navigation, while a graphics tablet can support precise tracing and drawing.

Digital art: A graphics tablet allows artists to control strokes with a stylus. Pressure-sensitive models can vary line thickness or brush effects according to stylus pressure.

> **Conclusion:** A mouse and trackball are useful for general navigation, while a graphics tablet is usually more suitable for precise drawing and pressure-sensitive digital art.

QUESTION 6 · EVALUATING

6. Plotters compared with an UHD monitor and large-format printer for blueprints

Introduction: Civil engineering blueprints require accurate lines, clear dimensions, appropriate scale and good print quality.

1. Drum plotter

A drum plotter moves paper around a rotating drum while a pen or drawing mechanism produces lines.

- Can produce detailed technical drawings and large-format output.
- Mechanical parts may require maintenance.
- Pen-based systems may be slower than modern printing systems; specialized equipment and consumables can increase operating costs.

2. Flatbed plotter

A flatbed plotter holds paper on a flat surface while the drawing mechanism moves over it.

- Produces precise drawings and keeps paper stationary during plotting.
- Suitable for specialized technical applications.
- Large machines require space, and paper size may be restricted by the plotting surface.

3. UHD monitor and large-format printer

- Engineers can zoom in and inspect fine details on screen.
- Digital drawings can be edited and stored easily.
- Large-format printing supports different paper sizes, depending on the printer.
- Print quality depends on the printer, media and settings; ink or toner and maintenance add operating costs.

Basis	Drum/flatbed plotter	UHD monitor + large-format printer

Line resolution	Depends on the plotter	Depends on printer resolution and settings
Scalability	Depends on machine design	Digital drawings can be resized; print size depends on printer
Editing	Often less convenient for digital revision	Easy to edit before printing
Storage	Physical drawings need storage space	Digital files are easy to store
Long-term cost	Depends on maintenance and consumables	Depends on ink/toner, media, maintenance and usage

> **Evaluation and conclusion:** For most modern civil engineering offices, an UHD monitor combined with a good-quality large-format printer is the better general choice because it supports convenient editing, digital storage and flexible printing. A specialized plotter may still be suitable where its particular capabilities are required. The final choice should depend on accuracy, paper size, workload and total cost of ownership.

QUESTION 7 · EVALUATING

7. Facial recognition compared with voice recognition in public, noisy environments

Introduction: Facial recognition and voice recognition are biometric technologies used to identify or authenticate people.

1. Facial recognition

Advantages: It can identify users without physical contact, does not depend on the user speaking and can be convenient for access control.

Limitations: Poor lighting, camera angle, masks and changes in appearance may affect accuracy. Photographs, videos or other presentation attacks may threaten security if anti-spoofing measures are weak. Public camera systems can raise privacy concerns.

2. Voice recognition

Advantages: It allows hands-free authentication, can be useful when touching a device is difficult and may support convenient remote access.

Limitations: High background noise can interfere with voice capture. Illness, stress and changes in speaking conditions may affect recognition. Recorded voices or synthetic speech may be used in attacks if safeguards are absent. Voice recordings can raise privacy concerns.

Factor	Facial recognition	Voice recognition
High ambient noise	Usually less affected by noise itself	Often affected by background noise
Poor lighting	May reduce accuracy	Not directly affected by lighting
Public exposure	Faces may be captured without cooperation	Voices may be recorded without permission
Security risks	Presentation attacks if poorly protected	Replay and synthetic-voice attacks if poorly protected
Privacy	Can enable unwanted visual tracking	Can expose sensitive voice recordings

Evaluation

In a high-noise public environment, facial recognition may be more reliable than voice recognition if lighting and camera positioning are suitable. However, neither technology is completely secure or reliable in every situation.

A strong system should use liveness detection, secure storage, encryption, access controls and a backup authentication method. High-risk systems may use multi-factor authentication rather than relying on a single biometric.

> **Conclusion:** Facial recognition is generally less affected by background noise, while voice recognition offers hands-free access. The safest choice depends on the environment, privacy requirements and security protections.

QUESTION 8 · CREATING

8. Design an accessible, hands-free computer workstation for a visually impaired user

Introduction: An accessible computer workstation enables a visually impaired user to operate a computer independently with assistive input and output technologies.

Components required

- **Computer:** A desktop or laptop with suitable accessibility software.
- **Voice recognition system:** Allows the user to control the computer, open applications and dictate text through spoken commands.
- **Screen reader:** Converts on-screen text and interface information into speech.
- **Voice response synthesizer:** Produces spoken output so the user can hear text, menus and notifications.
- **Tactile keyboard:** Provides raised or distinguishable keys that help users locate keys by touch. It is an optional backup input device rather than a fully hands-free solution.
- **Headphones or speakers:** Deliver clear audio feedback.
- **Braille display (optional):** Converts digital text into refreshable Braille for users who read Braille.

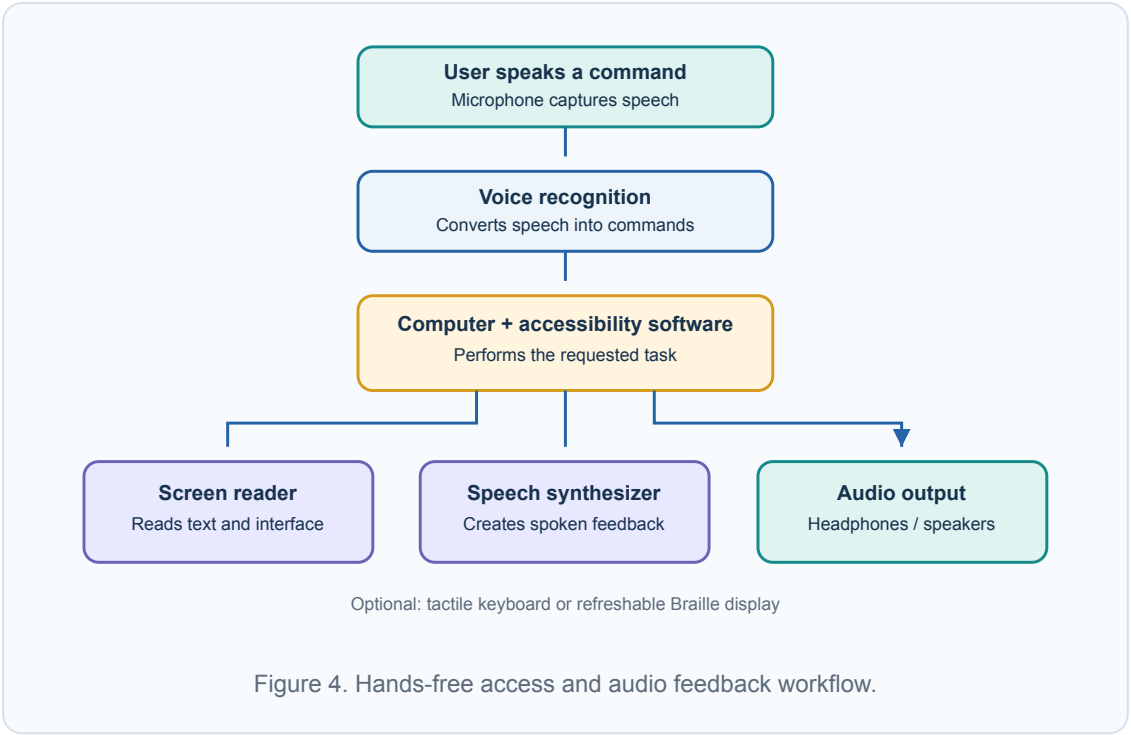


Figure 4. Hands-free access and audio feedback workflow.

Working of the workstation

1. The user gives a spoken command through a microphone.
2. The voice recognition system converts the command into a computer instruction.
3. The computer performs the requested task.
4. The screen reader reads relevant text and interface information aloud.
5. The speech synthesizer provides audio feedback through headphones or speakers.
6. If required, the user can use a tactile keyboard or refreshable Braille display for additional control and access.

Accessibility and safety features

- Provide adjustable speech speed and volume.
- Use keyboard shortcuts and clear spoken instructions.
- Ensure commonly used applications are compatible with screen readers.
- Provide an easy method to repeat or cancel commands.
- Offer privacy through headphones and secure user accounts.
- Arrange cables safely and keep the workstation accessible.

Advantages

- Improves independence and computer access.
- Reduces dependence on visual information.
- Supports writing, studying, communication and internet use.
- Makes digital learning more inclusive.
- Allows hands-free operation for many tasks.

> **Conclusion:** A workstation combining voice recognition, a screen reader and speech output can make computer use more accessible for a visually impaired user. Optional tactile and Braille devices provide additional ways to interact with digital information.

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